



RoadSOS+ — Product Requirement Document (PRD)

1. Product Overview

Product Name

RoadSOS+

Product Category

Emergency Road Assistance & Rescue Platform

Product Type

Location-Based Emergency Response Website / Progressive Web Application (PWA)

2. Product Purpose

RoadSOS+ is an intelligent emergency road assistance platform designed to help accident victims, bystanders, and responders quickly locate and contact nearby emergency and rescue services during road accidents.

The platform focuses on reducing delays during the **golden hour** by providing:

- Nearby trauma centres
- Ambulance services
- Police stations
- Towing and roadside assistance
- Mechanics and repair support
- Emergency SOS communication
- Offline emergency access in low-network areas

RoadSOS+ aims to simplify emergency response into a fast, accessible, and low-friction workflow.

3. Problem Statement

During road accidents, users often waste valuable time searching for:

- Nearby hospitals
- Ambulance numbers
- Police contacts
- Towing services
- Emergency support

This delay becomes critical during the golden hour after an accident.

In many regions:

- Emergency data is fragmented
- Internet connectivity may be weak
- Users panic and cannot navigate multiple apps
- Rescue support is difficult to coordinate quickly

RoadSOS+ solves this by providing a centralized emergency assistance platform.

4. Product Objectives

Primary Objectives

1. Instant Emergency Discovery

Allow users to quickly find nearby emergency and rescue services based on location.

2. Faster Emergency Response

Reduce time between accident occurrence and emergency action.

3. Offline Emergency Accessibility

Provide usable emergency support even in weak or unavailable network conditions.

4. Simplified SOS Workflow

Enable one-tap emergency actions during panic situations.

5. Scalable Emergency Infrastructure

Support future expansion across cities, states, and countries.

5. Success Metrics

User Metrics

- SOS trigger completion rate > 80%
- Nearby service load time < 5 seconds
- First emergency action within 10 seconds

Product Metrics

- Weekly active users
- Number of SOS activations
- Number of emergency searches
- Offline mode usage rate

Operational Metrics

- Emergency contact accuracy rate
- Successful location fetch percentage
- Cached/offline data reliability

Hackathon Evaluation Metrics

- Problem statement alignment
- Offline functionality
- Innovation score
- Practical implementation quality
- Scalability potential

6. User Personas

Persona 1 — Accident Victim

Profile

Driver or passenger involved in a road accident.

Needs

- Fast emergency assistance
- Minimal interaction effort

- Immediate SOS support

Pain Points

- Panic
- Injury
- Weak connectivity
- Lack of nearby service knowledge

Goal

Get help immediately.

Persona 2 — Bystander / Helper

Profile

Person assisting accident victims.

Needs

- Nearby hospital information
- Ambulance access
- Police contact
- Live location sharing

Pain Points

- Confusion during emergencies
- Difficulty finding correct services

Goal

Help victims quickly and correctly.

Persona 3 — Frequent Road User

Profile

Daily commuters, delivery riders, cab drivers, travelers.

Needs

- Reliable roadside assistance
- Mechanic and towing support

- Emergency preparedness

Goal

Stay protected during travel emergencies.

Persona 4 — Emergency Responder

Profile

Ambulance operators, volunteers, rescue personnel.

Needs

- Fast access to incident location
- Nearby rescue coordination
- Quick navigation support

Goal

Reach accident locations faster.

7. Core Features & Functionality

7.1 Emergency Service Finder

Description

Location-based discovery system for nearby emergency services.

Services Included

- Trauma centres
- Hospitals
- Ambulance services
- Police stations
- Fire stations
- Emergency helplines
- Towing services
- Mechanics
- Puncture repair shops

Functionalities

- GPS-based search
 - Radius filtering
 - Distance sorting
 - Map and list view
 - One-tap calling
 - Navigation support
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7.2 SOS Emergency Mode

Description

One-tap emergency assistance workflow.

Features

- Large SOS button
 - Emergency contact alerts
 - Live location sharing
 - SMS fallback alerts
 - Quick emergency call shortcuts
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7.3 Offline Emergency Support

Description

Emergency functionality during low or unavailable internet connectivity.

Features

- Cached emergency contacts
 - Offline maps
 - Last-known service lists
 - Local data storage
 - SMS fallback communication
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7.4 Vehicle Rescue Assistance

Description

Roadside vehicle rescue support.

Features

- Towing support discovery
 - Nearby mechanics
 - Puncture repair services
 - Breakdown assistance
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7.5 Emergency Contact Management

Features

- Save trusted contacts
 - Emergency quick-access list
 - Family notification setup
 - Region-wise emergency numbers
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8. Advanced / Optional Features

8.1 Road Safety Intelligence

Features

- Dangerous zone alerts
 - Accident-prone area indicators
 - Road hazard warnings
 - Pothole reporting
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8.2 Vehicle Health Assistance

Features

- Brake issue alerts
 - Overheating indicators
 - Fuel leak warnings
 - Vehicle status checks
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8.3 Multi-Region Expansion

Features

- Multi-language support
 - Country-wise emergency numbers
 - Localized service databases
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9. User Stories

Emergency User Stories

1.

As an accident victim, I want a one-tap SOS button so that I can request help immediately.

2.

As a bystander, I want to find nearby hospitals quickly so that I can help the injured person faster.

3.

As a user in a low-network area, I want cached emergency contacts so that I can still access support.

4.

As a commuter, I want towing and mechanic support so that I can recover from vehicle breakdowns.

5.

As a helper, I want to share my live location so that responders can reach the accident site easily.

Administrative User Stories

6.

As an admin, I want to update emergency contact databases so that users receive accurate information.

7.

As an admin, I want to organize services region-wise so that nearby results remain relevant.

8.

As an admin, I want outdated contacts flagged so that invalid emergency numbers are removed.

10. Website Modules

10.1 Home Page

Features

- SOS button
 - Emergency categories
 - Current location status
 - Nearby services preview
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10.2 Emergency Results Page

Features

- Nearby service listing
 - Distance information
 - Call buttons
 - Navigation support
 - Category filters
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10.3 SOS Page

Features

- Emergency trigger
 - Contact notifications
 - SMS alerts
 - Live location sharing
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10.4 Offline Emergency Page

Features

- Cached contacts
 - Offline emergency mode
 - Last updated data visibility
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10.5 Admin Dashboard

Features

- Emergency service management
 - Data verification
 - Regional categorization
 - Contact updates
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11. Technology Stack

Frontend

Choose one:

- React.js
 - Next.js
 - Flutter Web
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Backend

Choose one:

- Node.js + Express
 - Python Flask
 - FastAPI
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Database

Choose one:

- Firebase Firestore
 - MongoDB
 - PostgreSQL
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Maps & Location APIs

- Google Maps API
 - OpenStreetMap
 - Mapbox
 - Browser Geolocation API
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Notifications & Messaging

- Firebase Cloud Messaging
 - Twilio SMS API
 - Native Call Integration
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Offline Storage

- IndexedDB
 - SQLite
 - Local Cache
 - SharedPreferences / Hive (if app version)
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AI / Safety Layer (Optional)

- TensorFlow
 - PyTorch
 - OpenCV
 - YOLO
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12. Recommended Programming Languages

- JavaScript
 - TypeScript
 - Python
 - SQL
 - Dart (optional for Flutter)
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13. Proposed System Architecture

Layer 1 — User Interface

- SOS button
 - Service search
 - Emergency categories
 - Navigation controls
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Layer 2 — Location Layer

- GPS access
 - Location tracking
 - Radius search
 - Offline location fallback
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Layer 3 — Emergency Data Layer

- Hospitals
- Ambulances
- Police stations

- Mechanics
 - Towing services
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Layer 4 — Emergency Response Layer

- Calling system
 - SMS alerts
 - Live location sharing
 - Navigation launch
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Layer 5 — Intelligence Layer

- Hazard alerts
 - Accident warnings
 - Vehicle health monitoring
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14. MVP Scope (Hackathon Stage 1)

Minimum Viable Product Features

- SOS button
 - Nearby hospitals
 - Ambulance access
 - Police station finder
 - Towing support
 - Mechanic support
 - Offline emergency fallback
 - GPS location fetch
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15. Future Enhancements

Planned Enhancements

- AI hazard detection
- Pothole detection
- Smart rescue coordination
- Vehicle diagnostics
- Multi-country deployment

- Voice-based SOS activation
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16. Roadmap Document (Strategic Plan)

Phase 0 — Planning & Research

Goals

- Define MVP
 - Identify data sources
 - Finalize stack
 - Confirm architecture
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Phase 1 — MVP Development

Deliverables

- SOS workflow
 - Emergency search system
 - Nearby service discovery
 - Offline fallback
 - Emergency contact support
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Phase 2 — Reliability Improvement

Deliverables

- Verified contact database
 - Better ranking algorithm
 - Data refresh mechanisms
 - Improved response speed
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Phase 3 — Offline Optimization

Deliverables

- Enhanced caching
 - Lightweight emergency pages
 - SMS-first workflows
 - Better low-network support
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Phase 4 — Intelligence Layer

Deliverables

- Hazard alerts
 - Road risk indicators
 - AI-powered warnings
 - Safety analytics
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Phase 5 — Global Expansion

Deliverables

- Multi-language support
 - Country-specific emergency systems
 - Regional databases
 - International deployment readiness
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17. Non-Goals

The MVP will NOT initially include:

- Full autonomous accident detection
 - Advanced vehicle telemetry
 - Social networking features
 - Complex user profile systems
 - Heavy AI infrastructure
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18. Strategic Positioning

Product Positioning Statement

RoadSOS+ is a fast, location-aware emergency assistance platform that helps users instantly discover and contact nearby rescue services during road accidents, even in low-network environments.

19. Final Summary

RoadSOS+ is a practical and scalable emergency response platform focused on solving one of the most critical challenges during road accidents — finding help quickly.

The platform prioritizes:

- Fast emergency response
- Location-based rescue discovery
- Offline accessibility
- Simplified SOS workflows
- Real-world usability

RoadSOS+ strongly aligns with the IIT Madras Road Safety Hackathon theme and provides a feasible yet impactful emergency assistance solution.
